

Maths Class:

Mathematics

Term 3 Examination

TIME ALLOWED: 70 minutes

- Write your name and class at the top of this page.
- At the end of the examination this examination paper must be attached to the front of your answers.
- All questions may be attempted.
- All necessary working must be shown. Marks will be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- **START EACH QUESTION ON A NEW PAGE**

[illegible]

QUESTION 1: (9 marks)

(a) Fully factorise:

1 (i) $8x^3 - 1$

2 (ii) $y^4 - 16$

1 (b) Fully simplify $3\sqrt{2} \times 2\sqrt{6}$

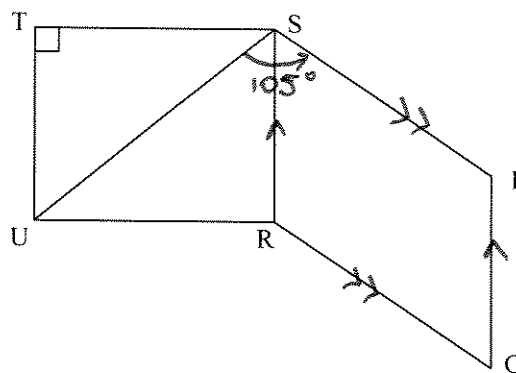
3 (c) If $(2 + 3\sqrt{2})^2 = a + \sqrt{b}$ find the values of a and b

2 (d) Fully simplify

$$\frac{x^2 - 64}{4x - 32}$$

QUESTION 2: (8 marks)

- 3 (a) STUR is a square, and
- $\angle USP = 105^\circ$
- .

Showing all reasoning, find $\angle PQR$ 

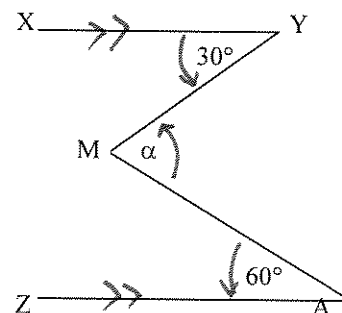
- 2 (b) Solve the equation

$$x^2 - 6x - 16 = 0$$

- 3 (c) (i) Copy the diagram at right onto your answer page

In the diagram, $\angle XYM = 30^\circ$, $\angle MAZ = 60^\circ$, and $XY \parallel ZA$

- (ii) By drawing a suitable line LN through M (or otherwise) and by labelling it appropriately, find the value of
- α
- , showing all reasoning



QUESTION 3: (9 marks)

- 2 (a) Solve

$$8 - x \geq -4$$

and graph the solution on a number line

- 2 (b) Find exact values for:

(i) $\sin 240^\circ$

(ii) $\sec (-45^\circ)$

- 2 (c) Rationalize the denominator of and simplify:

$$\frac{5}{\sqrt{6}-1}$$

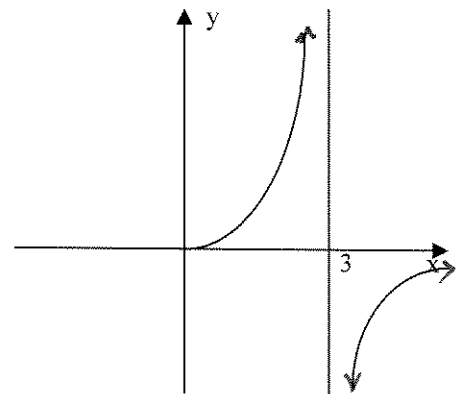
- 3 (d) Solve the following equations simultaneously:

$$\begin{cases} 5x - y - 4 = 0 \\ y = x^2 \end{cases}$$

QUESTION 4: (9 marks)

- 3 (a) Part of the graph of $y=g(x)$ is shown

Copy the diagram onto your answer sheet and complete the graph of $y=g(x)$ given that $g(x)$ is an EVEN function



(b) $H(x) = \sqrt{25 - x^2}$

- 1 (i) Sketch $y = H(x)$

- 2 (ii) What is the Domain and Range of the function $y = H(x)$?

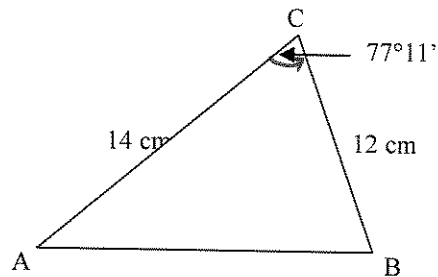
(c) If $f(x) = x^2 + 3$ and $h(x) = 2x - 1$

- 1 find (i) $f(-1)$

- 2 (ii) $h(f(y))$

QUESTION 5: (9 marks)

- 2 (a) Find the area of $\triangle ABC$, correct to 2 decimal places:



- 2 (b) Solve, for $0^\circ \leq \theta \leq 360^\circ$, the equation $\sin \theta = -\frac{1}{2}$

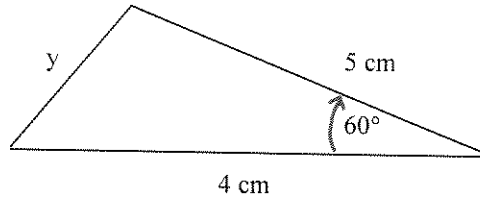
- (c) A ship sails from port P on a bearing of 15° , and after travelling 10km to Q, steers a new course of 105° , until it is due east of P (at the point R).

- 1 (i) Draw a diagram showing all the above information and labelling all points
- 2 (ii) Find, and label, all internal angles on your diagram
- 2 (iii) Find the distance from R to P, to the nearest 0.1 km

QUESTION 6: (9 marks)

- 2 (a) If $\tan \theta = -\frac{5}{12}$ and $90^\circ \leq \theta \leq 270^\circ$, find the exact value of $\cos \theta$ as a fraction.

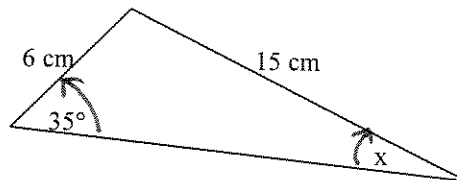
- 3 (c) Find the exact value of y in the following:



- 4 (b) Solve $2 \sin^2 x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$

QUESTION 7: (9 marks)

- 2 (a) In the diagram below, find the value of x correct to the nearest degree:



- 3 (b) Shade the region, in the first quadrant only, enclosed by the simultaneous inequalities:

$$x^2 + y^2 \leq 9$$

$$y < x$$

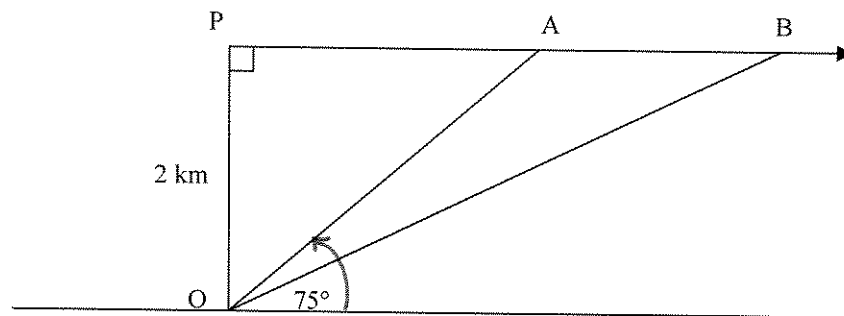
- 4 (c) Prove the following, showing all necessary lines of working:

$$\frac{\sin \theta}{1 - \cos \theta} + \frac{1 - \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$$

QUESTION 8: (8 Marks)

3 (a) If $\cos(x + 30) = \sin 2x$, find x

5 (b)



A plane flying horizontally at a constant speed over level ground at a height of 2 Km above the earth passes directly above a viewer on the ground at O . 1 minute later the observer finds the angle of elevation of the plane (at A) is 75° as shown. What will be the angle of elevation of the plane 1 minute later (at B)?

End of Examination

① (a) i. $(2x-1)(4x^2+2x+1)$ ①
 ii. $(y^2-4)(y^2+4)$ ①
 $= (y+2)(y-2)(y^2+4)$ ①

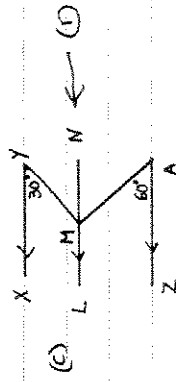
(b) $6\sqrt{12} = 12\sqrt{3}$ ①
 (c) $4+12\sqrt{2}+18 = 22+12\sqrt{2}$ ①
 $a = 22$ ① & $b = 288$ ①

(d) $\frac{(x+8)(x-8)}{4(x-8)} = \frac{x+8}{4}$ ①

② (a) $\angle USR = 45^\circ$ } ① for both
 (diagonals of a square bisect
 angles through which they pass)

$\angle RSP = 60^\circ$ ①
 $\therefore \angle PQR = 60^\circ$ ①
 (opposite angles of a parallelogram)

(b) $x^2 - 6x - 16 = 0$
 $(x-8)(x+2) = 0$
 $\therefore x = 8, -2$ ① and ①



$\angle YMN = 30^\circ$ (alternate angles)
 $XY \parallel LN$
 $\angle NMA = 60^\circ$ (alternate angles)
 $ZA \parallel LN$

$\therefore \alpha = 90$

③ (a) $8 - \bullet \geq -4$
 $-x \geq -12$
 $x \leq 12$ ①

must be completely correct.

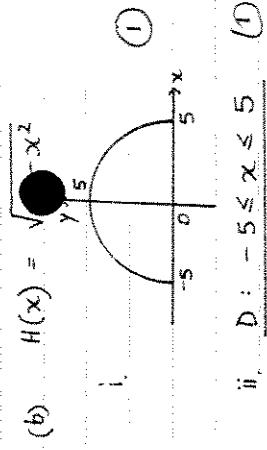
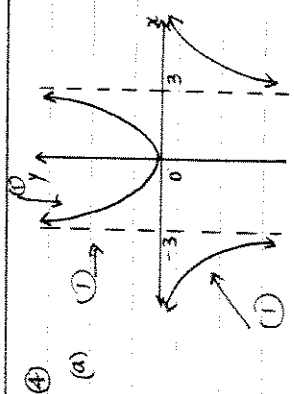
(b) i. $\sin 240^\circ = -\sin 60^\circ$
 $= -\frac{\sqrt{3}}{2}$ ①

ii. $\sec(-45^\circ) = \sec 315^\circ$
 $= \frac{1}{\cos 45^\circ}$
 $= \sqrt{2}$ ①

(c) $\frac{5}{\sqrt{6}-1} \times \frac{\sqrt{6}+1}{\sqrt{6}+1} = \frac{5\sqrt{6}+5}{6-1}$
 $= \frac{\sqrt{6}+1}{1}$ ①

(d) $5x - y - 4 = 0$
 $y = x^2$
 $5x - x^2 - 4 = 0$
 $x^2 - 5x + 4 = 0$ ①
 $(x-4)(x-1) = 0$

$\therefore x = 4, y = 16$ or ②
 $x = 1, y = 1$



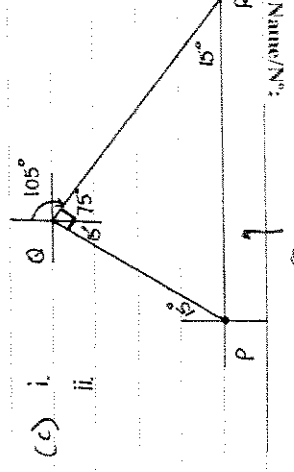
ii. $D: -5 \leq x \leq 5$ ①
 $R: 0 \leq y \leq 5$ ①

(c) $f(x) = x^2 + 3$
 $h(x) = 2x - 1$
 i. $f(-1) = (-1)^2 + 3$
 $= 4$ ①

ii. $f(y) = y^2 + 3$ ①
 $h(f(y)) = 2(y^2 + 3) - 1$
 $= 2y^2 + 6 - 1$
 $= 2y^2 + 5$ ①

⑤ (a) $A = \frac{1}{2} \times 14 \times 12 \times \sin 77^\circ$ ①
 $= 81.91$ ①

(b) $\sin \theta = -\frac{1}{2}$
 $\theta_{acute} = 30^\circ$
 $\therefore \theta = 210^\circ, 330^\circ$ ② each.



① for degree
 ① for the 15° , ① for the 75°

iii. $\cos 75^\circ = \frac{10}{PR}$ ①
 $PR = \frac{10}{\cos 75^\circ}$
 $\therefore PR = 38.6 \text{ km}$ ①

⑥ (a) $\tan \theta = -\frac{5}{12}$

$\cos \theta = -\frac{12}{13}$ ① for $\sin$
 ① for $\frac{12}{13}$

(b) $2 \sin^2 x - 1 = 0$
 $\sin^2 x = \frac{1}{2}$ ①
 $\sin x = \pm \frac{1}{\sqrt{2}}$
 $x_{acute} = 45^\circ$ ①
 $\therefore x = 45, 135, 225, 315$ ①

(c) $y^2 = 4^2 + 5^2 - 2 \times 4 \times 5 \times \cos 60^\circ$
 $y^2 = 41 - 20$ ①
 $\therefore y = \sqrt{21}$ ①

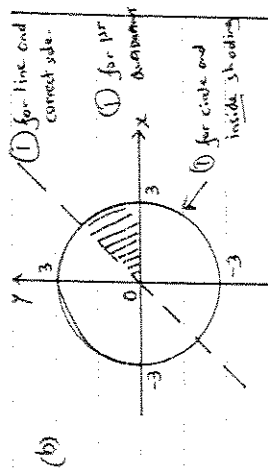
⑦ (a) $\frac{6}{\sin x} = \frac{15}{\sin 35^\circ}$ ①
 $\sin x = \frac{6 \sin 35^\circ}{15}$
 $\therefore x = 13^\circ$ ①

Teacher's Name: _____

Student's Name: _____

Teacher's Name: _____

Student's Name: _____



$$c) \frac{\sin \theta}{1 - \cos \theta} + \frac{1 - \cos \theta}{\sin \theta} = 2 \operatorname{cosec} \theta$$

$$\text{LHS} = \frac{\sin^2 \theta + (1 - \cos \theta)^2}{\sin \theta}$$

$$\begin{aligned} \text{for comm.} &\rightarrow \frac{\sin \theta (1 - \cos \theta)}{\sin \theta} \\ \text{den.} &= \frac{\sin^2 \theta + 1 - 2 \cos \theta + \cos^2 \theta}{\sin \theta} \end{aligned}$$

$$\text{for } \left(\frac{\sin \theta (1 - \cos \theta)}{\sin \theta} \right)$$

$$\text{Pythagoras} \rightarrow \frac{1 + 1 - 2 \cos \theta}{\sin \theta (1 - \cos \theta)}$$

$$= \frac{2 - 2 \cos \theta}{\sin \theta (1 - \cos \theta)}$$

$$= \frac{2(1 - \cos \theta)}{\sin \theta (1 - \cos \theta)}$$

$$\text{for cancellation} = \frac{2}{\sin \theta}$$

$$= 2 \operatorname{cosec} \theta$$

$$= \text{RHS}$$

$$\text{①} \downarrow$$

$$\text{⑧} \quad \downarrow$$

$$\text{⑩} \quad x + 30 + 2x = 90$$

$$3x = 60$$

$$x = 20$$

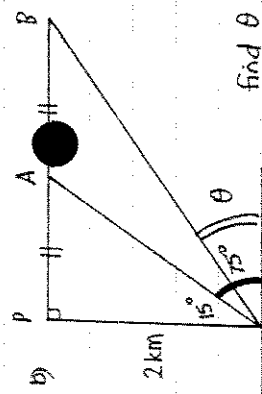
$$\text{or/}$$

$$\cos(x + 30) = \cos(90 - 2x)$$

$$x + 30 = 90 - 2x$$

$$3x = 60$$

$$x = 20$$



$$\tan 15^\circ = \frac{PA}{2}$$

$$PA = 2 \tan 15^\circ$$

$$PB = 2 \times 2 \tan 15^\circ$$

$$= 4 \tan 15^\circ$$

$$\tan \theta = \frac{2}{4 \tan 15^\circ}$$

$$\therefore \theta = 61^\circ 49'$$

$$\text{or/}$$

$$AP = x$$

$$\frac{2}{x} = \tan 75^\circ$$

$$\therefore x = \frac{2}{\tan 75^\circ}$$

$$\therefore x \approx 0.5359$$

$$\therefore PB = 1.0718$$

$$\tan \theta = \frac{2}{1.0718}$$

$$\therefore \theta = 61^\circ 49'$$